

Technical Assignment: Multi-Agent System for Long Document Intelligence (No Retrieval)

1. Overview

You are required to design and implement a multi-agent system capable of processing and reasoning over a large ($\approx 500+$ page) scanned document. The system must operate under strict constraints on context size, model choice, and architecture design, demonstrating the ability to handle long-form content without relying on retrieval-based methods (No RAG).

2. Objectives

Build a system that can:

- Convert a scanned document into machine-readable text.
- Process and understand large-scale content under context limitations.
- Support complex user queries requiring summarization, comparison, multi-section reasoning, and structured information extraction.

3. Constraints

3.1 Model & Data Constraints

- **Model:** Use only open-source models.
- **Context Window:** Maximum usable context window of 128K tokens.
- **Input:** $\sim 500+$ pages .

3.2 Architectural Constraints

- The system must follow a **multi-agent design** with clearly defined responsibilities for each agent.
- **No Retrieval:** The system must **not** use vector databases, similarity search, or embedding-based retrieval.

3.3 Processing Constraints

- Must handle OCR noise.
- No single step in the pipeline should exceed the 128K context constraint.

4. Functional Requirements

4.1 Document Ingestion & Processing

The system must extract text using OCR and break the document into smaller logical units

(sections, headings) to enable downstream agents to process these units independently while preserving structure.

4.2 Multi-Agent System Structure

The system must include multiple interacting agents that collectively achieve:

- **Local Understanding:** Analyzing specific document segments.
- **Aggregation:** Synthesizing local data into higher-level representations.
- **Global Reasoning:** Performing analysis over the entire document scope.

4.3 Context Management

Demonstrate a strategy for progressive information reduction or transformation to ensure all operations stay within the 128K limit while minimizing critical information loss.

4.4 Query Interface

Support for:

- Full and section-level summarization.
- Comparison across distant sections.
- Extraction of entities, risks, and decisions.
- Identification of inconsistencies or contradictions.

5. Deliverables

Deliverable	Description
Source Code	Fully functional, modular implementation.
Documentation	README including architecture diagrams, agent roles, data flow, and design tradeoffs.
Demonstration	Examples of processing a large document with 3 complex query types.

6. Evaluation Criteria

- **Architecture:** Clarity of multi-agent design.
- **Context Handling:** Efficiency in managing large inputs within 128K tokens.
- **Reasoning:** Quality of grounded, multi-step understanding.
- **Robustness:** Ability to handle noisy OCR and complex structures.

Note: This assignment evaluates system design under constraints and the ability to build scalable, reasoning-oriented pipelines. Focus on design quality and tradeoff awareness.